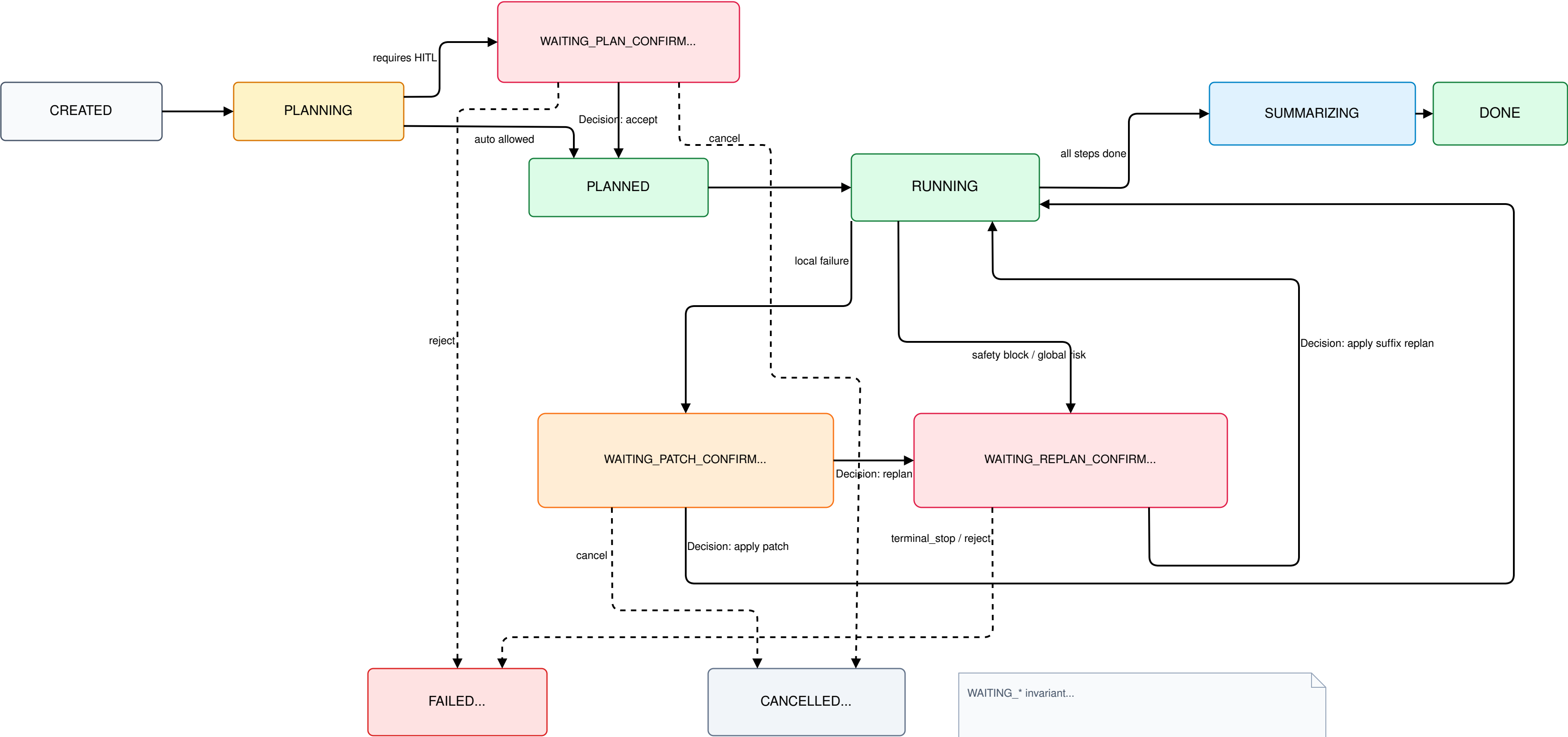


FSM 状态转移图：外部生命周期、内部恢复态与 HITL 决策边界

状态突变由 Workflow/FSM 统一负责；WAITING_* 表示执行暂停，终态不可再修改。



WAITING_* invariant...

ExternalStatus exposes WAITING_PATCH_CONFIRM / WAITING_REPLAN_CONFIRM; InternalStatus may use WAITING_PATCH, PATCHING, WAITING_REPLAN, REPLANNING for implementation detail.